

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Information Technology) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Information Technology) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTTTPE603A - SOFTWARE TESTING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 Which of the following is NOT a level of testing?
 1. Unit testing
 2. Integration testing
 3. Regression testing
 4. System testing
- 2 What do we mean by verification in software testing?
 1. Testing the actual software
 2. Evaluating software during development
 3. Fixing bugs
 4. Performing user acceptance tests
- 3 In bottom-up integration testing, what is commonly used to simulate higher-level modules?
 1. Stubs
 2. Drivers
 3. Simulators
 4. Mocks
- 4 Which of the following best characterizes functional testing?
 1. Evaluating system security under stress
 2. Measuring response time under load
 3. Verifying what the system does based on specifications
 4. Testing usability and user interface
- 5 Which non-functional testing type checks the system's ability to handle maximum user load?
 1. Stress testing
 2. Functional testing
 3. Regression testing
 4. Sanity testing

- 6 Which of the following is MOST critical in Internationalization testing?
 1. Unit testing
 2. Locale data formatting (dates, numbers, currencies)
 3. Load testing
 4. Functional coverage
- 7 Which of the following is typically tested in Internationalization but NOT Localization?
 1. Text expansion due to translation
 2. UI layout adaptation
 3. Separation of UI strings from code logic
 4. Accuracy of translated content
- 8 Which performance metric is most relevant for end-user experience?
 1. CPU usage
 2. Response time
 3. Memory leaks
 4. Database hit ratio
- 9 What is the significance of higher cyclomatic complexity in a program?
 1. Easier to maintain and test
 2. Indicates better performance
 3. Code has more independent paths and is harder to test
 4. Code will always be faster
- 10 In object-oriented testing, which of the following focuses on class behavior as a black box?
 1. State-based testing
 2. White-box testing
 3. Mutation testing
 4. Path testing
- 11 Use case-based testing is most effective when applied at which testing level?
 1. Unit testing
 2. System and acceptance testing
 3. Statement-level testing
 4. Regression testing only
- 12 In a Defect Bash, who typically participates in testing?
 1. Only QA testers
 2. Developers only
 3. Testers, developers, product owners, business analysts, and sometimes end users
 4. Only project managers

Q2. Solve the following.

- A) Using examples from software development, demonstrate how quality assurance activities differ from quality control activities in practice. 6
- B) Differentiate between white box testing (WBT) and Black box testing with its techniques. 6

- Q3. Solve the following.
- A) Describe how you would plan and execute a defect bash for a mobile app release, including which team members should participate and why 6
 - B) Differentiate between stubs and drivers in top-down and bottom-up integration testing? 6
- Q4. Solve Any Two of the following.
- A) For each scenario, identify whether it describes Functional or Non-functional testing: 6
 - 1. Verifying that a login screen properly authenticates users with valid credentials.
 - 2. Measuring how quickly a website responds to 1,000 simultaneous users.
 - 3. Checking if an e-commerce checkout process calculates discounts correctly.
 - 4. Assessing whether error messages are clear and helpful to users.
 - 5. Confirming a banking app encrypts all sensitive data transmissions.
 - 6. Testing if a flight booking system correctly applies business-class upgrade rules.
 - B) Explain system testing and acceptance testing in detail 6
 - C) Differentiate between functional testing and non-functional testing 6
- Q5. Solve Any Two of the following.
- A) Explain the need of Internationalization and discuss it in detail 6
 - B) Explain the purpose and importance of performance testing and regression testing in the software development life cycle. How do these testing types help ensure the reliability and stability of software applications? 6
 - C) Differentiate between Error Seeding and mutation testing 6
- Q6. Solve Any Two of the following.
- A) Differentiate between Procedural and Object-Oriented software testing 6
 - B) Explain State, Attribute and Category based partition testing in object-oriented testing 6
 - C) Explain any two tools used in for testing object oriented software. 6

*** End ***